

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Winter 2023

Course: B. Tech. Branch: Civil Engineering

Semester: VI Semester

Subject Code & Name: BTCVOE605H Applications of Remote Sensing and GIS

Max Marks: 60

Date: 25/01/2024

Duration: 3.00 Hr.

/Instructions to the Students:/

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A)** Write short notes on: - **6**
- (a) Remote sensing platforms
 - (b) Atmospheric windows
 - (c) Electromagnetic radiation
- B)** Write a detailed note on atmospheric scattering and its types. **6**
- C)** Explain in detail about methods of remote sensing. **6**

Q.2 Solve Any Two of the following.

- A)** Write short notes on: - **6**
- (a) Earth observation satellite
 - (b) Swath
 - (c) RADAR
- B)** Write detailed note on spectral, spatial and temporal resolution. **6**
- C)** Write a detailed note on microwave remote sensing technique and its applications. **6**

Q. 3 Solve Any Two of the following.

- A)** Write short notes on: - **6**
- (a) Supervised classification of satellite imagery
 - (b) Spectral signature of soil
 - (c) Digital image
- B)** Write a detailed note on the steps involved in digital image processing. **6**
- C)** What are important keys in image interpretation and their importance? **6**

Q.4 Solve Any Two of the following.

- A)** Write short notes on: - **6**
- (a) Digital Elevation Model
 - (b) Cloud computing
 - (c) Triangular Irregular Network (TIN)
- B)** Explain the working of Light Detection and Ranging (LiDAR) system with its applications. **6**
- C)** Define and describe the Geographic Information System with its working. **6**

Q. 5 Solve Any Two of the following.

- A)** Write short notes on: - **6**
- (a) Geoid
 - (b) Approximation of Earth
 - (c) Universal Transverse Mercator system
- B)** Explain in detail about datum system and describe its types. **6**
- C)** Explain Lambert Conformal Conic Projection system and its advantages and disadvantages. **6**

***** End *****